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Disposable paper pulp fast food cutlery

Abstract

A piece of pulp cutlery includes a handle, a working body, and a transition portion by which the handle is connected with the working body. The handle includes two curved lateral edge portions, and a first double arch reinforcing structure disposed between the two curved lateral edge portions and extending along a length of the handle. The transition portion includes a second double arch reinforcing structure which extends from the first double arch reinforcing structure and toward the working body. The working body includes a third double arch reinforcing structure which extends from the second double arch reinforcing structure and toward a lowest region of the working body.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to cutlery made from a biodegradable pulp material and in particular to a disposable paper pulp fast food fork, a disposable paper pulp fast food spoon, and a disposable paper pulp fast food knife.
BACKGROUND
(2) Disposable cutlery is widely applied in the catering field. Along with development of fast food, tourism and take-out services, the use amount of the disposable cutlery increases dramatically. In the prior art, most of the disposable cutlery are made of plastic or wood. The plastic cutlery may take dozens of years or even several hundred years to degrade, bringing environmental pollution, while the production of the wooden cutlery requires a large number of trees to be felled, which is also unfavorable for environmental improvement.
(3) In recent years, as people have increasing awareness for environmental protection, more and more efforts are made to develop disposable cutlery using a degradable material such as paper pulp or pulp. However, in the prior art, the disposable paper pulp or pulp cutlery generally has a low strength such that they are easy to deform or break in use. Furthermore, in the prior art, some of the disposable paper pulp cutlery are thickened to have an increased strength. In this case, the inconvenience of use is brought, the comfort of use is affected and the production costs are increased.
(4) Chinese utility model patent ZL201922474635.6 discloses a paper-made cutlery and a paper-made cutlery set, and specifically discloses the following: a strength of the cutlery is increased by forming a groove on a holding portion and disposing a reinforcing rib on a connection portion. In use, a functional portion of the cutlery conveys a load to the holding portion along a center axis of the cutlery (i.e., a fork, a spoon, or a knife) after receiving the load, and the middle portion of the holding portion is a thin-walled groove structure, and thus only a flange at the top of the groove is

incapable of providing effective support for conveying the load due to a low strength especially in a case of thin wall, thereby bending or twisting the cutlery, and further affecting effective use of the cutlery. Furthermore, the top of the groove has a larger width than the bottom of the groove, and the larger width of the groove means that the strength of the handle at this position is lower. In addition, because the holding portion is recessed in middle and bulged at both sides, stability and comfort in use are affected to some extent.

SUMMARY

(5) The present disclosure aims to provide a disposable paper pulp or pulp fast food fork, where a strength of a handle of the fast food fork can be increased through an undulated curved surface structure, and an integral thin-walled structure of the fast food fork is formed using paper pulp, thereby reducing the production costs and increasing the use comfort; to provide a disposable paper pulp fast food spoon, which can still have good structural strength in a case of small material thickness through disposal of a reinforcing structure, thereby ensuring use comfort of the fast food spoon; and to provide a disposable paper pulp fast food knife, where the entire structural strength of the fast food knife can be increased by a reinforcing structure, thereby ensuring effective use of a thin paper pulp fast food knife.

(6) An aspect of the present disclosure is achieved in the following technical solutions.

(7) Provided is a disposable paper pulp fast food fork, including a handle, a first transition portion and a fork body. An end of the handle is connected to an end of the fork body through the first transition portion to form a fast food fork body, and a plurality of prongs is formed at the other end of the fork body.

(8) Two reinforcing arched strips paralleled to each other along a length direction of the handle are convexly disposed on an upper surface of the handle, and a first reinforcing groove lower than upper surfaces of the reinforcing arched strips is formed between the two reinforcing arched strips; one end of the reinforcing arched strip extends to an end of the handle away from the first transition portion, and the other end of the reinforcing arched strip extends through the first transition portion to the lowest region of the fork body; a second reinforcing groove is formed on each prong, one end of the second reinforcing groove extends to a tip of the prong, and the other end of the second reinforcing groove extends through the lowest region of the fork body to the first transition portion; the reinforcing arched strips, the first reinforcing groove, the prongs and the second reinforcing grooves are disposed in parallel.

(9) The second reinforcing groove on the prong located in middle extends into the first reinforcing groove, and the second reinforcing grooves on several prongs located at both sides of the middle prong are symmetrically distributed at both sides of an end of the first reinforcing groove.

(10) A reinforcing concave surface is formed at a middle portion of an upper surface of the fork body, and the reinforcing concave surface is located at the lowest region of the fork body and in smooth transitional connection with the two reinforcing arched strips.

(11) A third transition portion is formed at each of the ends of the two reinforcing arched strips close to the first transition portion and the reinforcing arched strips are in smooth transitional connection with the reinforcing concave surface of the fork body through the third transition portions.

(12) A downwardly-recessed third reinforcing groove is formed on the upper surface of the handle, one end of the third reinforcing groove extends to an end of the handle away from the first transition portion, and the other end of the third reinforcing groove extends through the first transition portion to the lowest region of the fork body, such that the two reinforcing arched strips are disposed within the third reinforcing grooves.

(13) A second transition portion is formed at each of ends of the two reinforcing arched strips, and the two reinforcing arched strips are in smooth transitional connection with the handle through the second transition portions.

(14) The two reinforcing arched strips are connected with each other through a connection bridge.

(15) The two reinforcing arched strips are connected with each other through a fourth transition portion, the fourth transition portion is a convex arc-surface structure and located at an end of the handle away from the first transition portion, and the fourth transition portion is connected smoothly between the two reinforcing arched strips and the third reinforcing groove of the handle.
(16) The upper surfaces of the two reinforcing arched strips both are inclined surface structures with an angle of inclination being 1.4°-1.6°, a high end of the inclined surface structure is located at an end of the handle away from the first transition portion, and a low end of the inclined surface structure is located at an end of the handle connected with the first transition portion.
(17) The fast food fork body is an integral thin-walled structure with a thickness of 0.6-0.9 mm.
(18) Another aspect of the present disclosure is achieved in the following technical solutions.
(19) Provided is a disposable paper pulp or pulp fast food spoon, including a handle, a transition portion, and a spoon body. An end of the handle is connected with an end of the spoon body through the transition portion to form a fast food spoon body.
(20) A U-shaped recess is formed on an upper surface of the handle, and the U-shaped recess extends from an end of the handle to the transition portion; a double arch reinforcing portion is disposed in the U-shaped recess, and an upper surface of the double arch reinforcing portion is higher than the handle; the double arch reinforcing portion is formed by two paralleled arched strips which are disposed along edges of two sides of the U-shaped recess respectively, a nesting groove is formed between the two arched strips, and an undulant curved surface structure is formed between the two arched strips, the nesting groove, the U-shaped recess and the handle; an end of the arched strip extends to an end of the U-shaped recess and the other end of the arched strip extends toward the lowest region of the spoon body through the transition portion.
(21) The nesting groove is disposed along a center axis of the handle, a connection bridge is disposed in the nesting groove, and one or more connection bridges are connected between the two arched strips respectively.
(22) An upper surface of the arched strip is an inclined surface structure with an angle of inclination being 1.0°-1.6°, an inclined surface high end of the arched strip is located at an end of the handle away from the transition portion, and an inclined surface low end of the arched strip is located at an end of the handle close to the transition portion.
(23) A first smooth transition portion is formed at the inclined surface high end of the arched strip, and the inclined surface high end of the arched strip is in smooth transitional connection with an end of the handle through the first smooth transition portion.
(24) A second smooth transition portion is formed at the inclined surface low end of the arched strip and is in smooth transitional connection with the spoon body through the transition portion.
(25) A third smooth transition portion is formed at an end of the double arch reinforcing portion, and the inclined surface high ends of the two arched strip are in smooth transitional connection with an end of the handle through the third smooth transition portion.
(26) The handle, the transition portion, the spoon body and the double arch reinforcing portion are formed into an integral thin-walled fast food spoon through paper pulp solidification.
(27) The upper surface of the handle, an upper surface of the transition portion and an upper surface of the spoon body are located in a same plane; an end of the U-shaped recess of the handle is in smooth transitional connection with the upper surface of the handle, the other end of the U-shaped recess is in smooth transitional connection with the transition potion, and the transition portion is in smooth transitional connection with the spoon body.
(28) The integral thin-walled fast food spoon has a thickness of 0.6-0.9 mm
(29) Still another aspect of the present disclosure is achieved in the following technical solutions.
(30) Provided is a paper pulp or pulp fast food knife including a handle and a knife body. The knife body is connected to an end of the handle, and the handle and the knife body both are located in a same plane to form a fast food knife body.
(31) A downwardly-recessed first groove body is formed on an upper surface of the fast food knife

body and extends from an end of the handle to a middle portion of the knife body. Two knife body reinforcing ribs are convexly disposed in the first groove body and extend from an edge of an end of the handle to the knife body. The two knife body reinforcing ribs are disposed in parallel along a length direction of the handle, and a downwardly-recessed reinforcing groove is formed between the two knife body reinforcing ribs. A downwardly-recessed second groove body is formed on an upper surface of the knife body. One end of the second groove body is in smooth transitional connection with the first groove body, and the other end of the second groove body extends to a tip of the knife body along a length direction of the knife body.
(32) An end of the knife body reinforcing rib away from a cutting edge extends to a connection position of the first groove body and the second groove body.
(33) Upper surfaces of the two knife body reinforcing ribs are inclined surface structures higher than an upper surface of the handle. A high end of the inclined surface structure is located at an end of the handle, and a low end of the inclined surface structure is located on the knife body.
(34) The inclined surface structures of the two knife body reinforcing ribs have an angle of inclination of 1.4°-1.6°.
(35) A first transition portion is formed at the high ends of the inclined surface structures of the two knife body reinforcing ribs, and a second transition portion is formed at the low ends of the inclined surface structures of the two knife body reinforcing ribs. The knife body reinforcing ribs are in smooth transitional connection with the first groove body through the first transition portion, and the knife body reinforcing ribs are in smooth transitional connection with the knife body through the second transition portion.
(36) The first transition portion includes two strip-shaped first unit transition portions such that each of the knife body reinforcing ribs is connected with the first unit transition portion to form an elongated reinforcing rib parallel to the length direction of the handle.
(37) A connection bridge is disposed between the two elongated reinforcing ribs and located in the reinforcing groove.
(38) The first transition portion is an arc-surface structure such that the first transition portion is in smooth transitional connection with the two knife body reinforcing ribs and the first groove body.
(39) The second transition portion comprises two strip-shaped second unit transition portions such that each of the knife body reinforcing ribs is connected with the second unit transition portion to form an elongated reinforcing rib parallel to the length direction of the handle, wherein one elongated reinforcing rib is in smooth transitional connection with the first groove body and the other elongated reinforcing rib is in smooth transitional connection with the first groove body and the second groove body.
(40) The disposable paper pulp fast food knife is an integral thin-walled structure formed through paper pulp solidification, and has a thickness of 0.6-0.9 mm
(41) Compared with the prior art, the present disclosure has the following beneficial effects.
(42) 1. In an aspect of the present disclosure, two reinforcing arched strips are convexly disposed on the handle and the first reinforcing groove is formed between the two reinforcing arched strips, such that a bending strength of the handle can be increased by use of the two paralleled reinforcing arched strips, and bending strength and torsional strength of the handle both can be improved by use of the undulated curved surface structure. In this way, deformation of the handle of the fast food fork under a force is avoided and its nesting with the cutlery of same specifications is facilitated at the same time.
(43) 2. In an aspect of the present disclosure, the second reinforcing grooves are disposed on the prongs, the second reinforcing grooves are partially overlapped with the first reinforcing groove, and a reinforcing concave surface is disposed on the fork body. In this case, the strength of the fork body is increased and a load can be transferred easily from the fork body to the handle, thus improving the entire structural strength of the fast food fork and effectively avoiding the deformation of the fork body of the fast food fork under a force.

(44) 3. In an aspect of the present disclosure, the connection bridge, the second transition portion, the third transition portion and the fourth transition portion are disposed selectively on the reinforcing arched strips of the handle, such that the structural strength and the comfort of the handle are further improved. Further, information such as cavity position number can be printed conveniently.
(45) 4. In an aspect of the present disclosure, the fast food fork is formed integrally through paper pulp solidification. Thus, the fork is easy in mold opening and process operation, simple in structure, small in volume and light in weight and can be manufactured in mass production. As a result, the forks can be widely applied in fast food delivery.
(46) 5. In an aspect of the present disclosure, the strength of the handle of the fast food fork can be increased by the undulated curved surface structure formed of the handle, the third reinforcing groove, the reinforcing arched strips and the first reinforcing groove, and the structural strength of the fork body can be increased by the second reinforcing grooves and the reinforcing concave surface. Thus, an integral thin-walled structure of the fast food fork can be formed using paper pulp, thus reducing the production costs. Furthermore, the nesting packaging of the fast food forks is facilitated and the holding comfort of the fast food forks is guaranteed.
(47) 6. In another aspect of the present disclosure, the double arch reinforcing portion formed by two paralleled arched strips disposed along a length direction of the handle is provided to resist the bending. Further, the smooth transition portions further increase connection strength between the arched strips and the entire structure of the fast food spoon, thus effectively avoiding deformation of the fast food spoon resulted from a force.
(48) 7. In another aspect of the present disclosure, a connection bridge is formed between two arched strips to achieve the purpose of resisting a twist so as to further improve the strength of the handle and the holding comfort. Furthermore, mutual nesting of different spoons can be facilitated and information such as cavity position number can be printed.
(49) 8. In another aspect of the present disclosure, the undulant curved surface structure is formed by two arched strips, the U-shaped recess and the nesting groove to achieve symmetrical design. Thus, the handle has a higher strength than a handle having a plane structure and can effectively prevent deformations such as bending and twist. In this way, a thin fast food spoon can be manufactured while ensuring the structural strength, thereby reducing production cost, storage cost and transportation cost.
(50) 9. In another aspect of the present disclosure, due to integral paper pulp solidification, it is easy to perform mold opening and process operations, thus simplifying the structure and facilitating mass production.
(51) 10. In another aspect of the present disclosure, the disposal of the two parallel arched strips enables the fast food spoon to have good structural strength in a case of small material thickness. Furthermore, by use of the smooth transition portions at both ends of the arched strips and the nesting groove, the overall strength of the fast food spoon is further increased and the use comfort of the fast food spoon is guaranteed.
(52) 11. In still another aspect of the present disclosure, two knife body reinforcing ribs are disposed, the reinforcing groove is formed between the two knife body reinforcing ribs, and the two knife body reinforcing ribs are convexly disposed on the first groove body of the handle. In this case, an undulated curved surface structure is formed on the handle. Compared with a planar structure, the handle has higher strength, thus leading to better bending strength and better torsional strength. In this way, the effective use of the thin-walled paper pulp fast food knife is guaranteed.
(53) 12. In still another aspect of the present disclosure, due to disposal of the second groove body, the first groove body and two parallel knife body reinforcing ribs, a load on the knife body can be transferred along a direction of the handle. In this case, the fast food knife can receive a force uniformly, and deformation or the like occurring to the fast food knife in use can be avoided. The fast food knife can still work normally even in a case of small thickness.

(54) 13. In still another aspect of the present disclosure, with disposal of the reinforcing structures such as the transition portion and the connection bridge, the entire structural strength of the fast food knife can be further increased to facilitate transfer of a load and the external appearance of the fast food knife is made more beautiful.
(55) 14. In still another aspect of the present disclosure, with disposal of the knife body reinforcing ribs and the reinforcing groove, nesting packaging of the cutlery of the same specifications can be facilitated for ease of use. Further, the present disclosure is small in volume and light in weight, and thus can be easily promoted, thereby bringing good economic benefits.
(56) 15. In still another aspect of the present disclosure, the strength of the handle and the knife body can be increased by forming an undulated curved surface structure on the fast food knife, and a load can be transferred from the knife body to the handle through the first groove body, the second groove body and the knife body reinforcing ribs. With disposal of the reinforcing structures, the entire structural strength of the thin-walled paper pulp fast food knife can be increased and the holding comfort is improved, thereby ensuring the effective use of the fast food knife.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a disposable paper pulp fast food fork according to an embodiment 1-1 of the present disclosure.
(2) FIG. **2** is a front view of a disposable paper pulp fast food fork according to an embodiment 1-1 of the present disclosure.
(3) FIG. **3** is a side view of a disposable paper pulp fast food fork according to an embodiment 1-1 of the present disclosure.
(4) FIG. **4** is an enlarged sectional view taken along a line A-A in FIG. **2**.
(5) FIG. **5** is an enlarged sectional view taken along a line B-B in FIG. **2**.
(6) FIG. **6** is an enlarged sectional view taken along a line C-C in FIG. **2**.
(7) FIG. **7** is a perspective view of a disposable paper pulp fast food fork according to an embodiment 1-2 of the present disclosure.
(8) FIG. **8** is a front view of a disposable paper pulp fast food fork according to the embodiment 1-2 of the present disclosure.
(9) FIG. **9** is an enlarged sectional view taken along a line D-D in FIG. **8**.
(10) FIG. **10** is an enlarged sectional view taken along a line E-E in FIG. **8**.
(11) FIG. **11** is a stereoscopic diagram of a disposable paper pulp fast food spoon according to the present disclosure.
(12) FIG. **12** is a front view of a disposable paper pulp fast food spoon according to the present disclosure.
(13) FIG. **13** is an enlarged sectional view taken along A-A in FIG. **12**, wherein the disposable paper pulp fast food spoon is considered in an upright position.
(14) FIG. **14** is an enlarged sectional view taken along B-B in FIG. **12**, wherein the disposable paper pulp fast food spoon is considered in the upright position.
(15) FIG. **15** is a side view of a disposable paper pulp fast food spoon according to the present disclosure.
(16) FIG. **16** is a stereoscopic diagram of an embodiment 2-2 of a disposable paper pulp fast food spoon according to the present disclosure.
(17) FIG. **17** is a stereoscopic diagram of an embodiment 2-3 of a disposable paper pulp fast food spoon according to the present disclosure.
(18) FIG. **18** is a perspective view of a disposable paper pulp fast food knife according to an embodiment 3-1 of the present disclosure.

(19) FIG. **19** is a front view of a disposable paper pulp fast food knife according to the embodiment 3-1 of the present disclosure.
(20) FIG. **20** is a side view of a disposable paper pulp fast food knife according to an embodiment 3-1 of the present disclosure.
(21) FIG. **21** is an enlarged sectional view taken along a line A-A in FIG. **19**.
(22) FIG. **22** is an enlarged sectional view taken along a line B-B in FIG. **19**.
(23) FIG. **23** is a perspective view of a disposable paper pulp fast food knife according to an embodiment 3-2 of the present disclosure.
(24) FIG. **24** is a front view of a disposable paper pulp fast food knife according to the embodiment 3-2 of the present disclosure.
(25) FIG. **25** is an enlarged sectional view taken along a line C-C in FIG. **24**.
(26) FIG. **26** is an enlarged sectional view taken along a line D-D in FIG. **24**.
(27) Numerals of the drawings are described as follows: **101** handle, **111** reinforcing arched strip, **112** first reinforcing groove, **113** connection bridge, **114** third reinforcing groove, **115** second transition portion, **116** third transition portion, **117** fourth transition portion, **102** first transition portion, **103** fork body, **131** prong, **132** second reinforcing groove, **133** reinforcing concave surface, **201** handle, **211** U-shaped recess, **202** transition portion, **203** spoon body, **204** double arch reinforcing portion, **241** arched strip, **2411** first smooth transition portion, **2412** second smooth transition portion, **242** nesting groove, **243** connection bridge, **244** third smooth transition portion, **301** handle, **302** knife body, **303** first groove body, **304** knife body reinforcing rib, **341** reinforcing groove, **342** first transition portion, **343** second transition portion, **344** connection bridge, and **305** second groove body.
DETAILED DESCRIPTIONS OF EMBODIMENTS
(28) The present disclosure will be further described below in combination with the accompanying drawings and the specific embodiments.
(29) As shown in FIGS. **1** and **7**, provided is a disposable paper pulp fast food fork. The disposable paper pulp fast food fork includes a handle **101**, a first transition portion **102**, and a fork body **103**. An end of the handle **101** is connected to an end of the fork body **103** through the first transition portion **102** to form a fast food fork body, and a plurality of prongs **131** are formed at the other end of the fork body **103**.
(30) As shown in FIGS. **1-3**, **7** and **8**, two reinforcing arched strips **111** paralleled to each other along a length direction of the handle **101** are convexly disposed on an upper surface of the handle **101**, and a first reinforcing groove **112** lower than upper surfaces of the reinforcing arched strips **111** is formed between the two reinforcing arched strips **111** as shown in FIGS. **4** and **10**. An undulated curved surface structure is formed among the two reinforcing arched strips **111**, the first reinforcing groove **112** and the handle **101** and has a higher strength to increase a bending strength and a torsional strength of the handle **101** effectively as compared with a planar structure of the handle **101**. The inverse W-shaped structure formed by the two reinforcing arched strips **111** and the first reinforcing groove **112** can enable the handle **101** to have nesting function so as to facilitate nesting of fast food forks of the same specifications. One end of the reinforcing arched strip **111** extends to an end of the handle **101** away from the first transition portion **102**, and the other end of the reinforcing arched strip **111** extends through the first transition portion **102** to the lowest region of the fork body **103**. As clearly shown in FIG. **1**, for example, the two reinforcing arched strips **111** and the first reinforcing groove **112** form a double arch reinforcing structure or portion. More specifically, the handle **101** includes a first double arch reinforcing structure disposed between two curved lateral edge portions of the handle **101** and extending along a length of the handle **101**. The first transition portion **102** includes a second double arch reinforcing structure extending from the first double arch reinforcing structure and toward the fork body **103**. The fork body **103** includes a third double arch reinforcing structure extending from the second double arch reinforcing structure and toward a lowest region of the fork body **103**. The first double

arch reinforcing structure, the second double arch reinforcing structure, and the third double arch reinforcing structure form a single one double arch reinforcing structure. A second reinforcing groove **132** is formed on each prong **131**, one end of the second reinforcing groove **132** extends to a tip of the prong **131**, and the other end of the second reinforcing groove **132** extends through the lowest region of the fork body **103** to the first transition portion **102**. The disposal of the second reinforcing grooves **132** increases the strength of the prongs **131** to guarantee normal use of the prongs **131**. The reinforcing arched strips **111**, the first reinforcing groove **112**, the prongs **131** and the second reinforcing grooves **132** are disposed in parallel to facilitate transfer of a load and ensure the entire structural strength of the fast food fork body.

(31) The second reinforcing groove **132** on the prong **131** located in middle extends into the first reinforcing groove **112**, and the second reinforcing grooves **132** on several prongs **131** located at both sides of the middle prong **131** are symmetrically distributed at both sides of an end of the first reinforcing groove **112**. The prongs **131** transfers a load to the reinforcing arched strips **111** and the first reinforcing groove **112** through the second reinforcing grooves **132**, and the entire structural strength of the fork body **103** is guaranteed by use of the symmetrically-arranged second reinforcing grooves **132**. In this way, the use process of the fork body **103** is guaranteed.

(32) As shown in FIGS. **1**, **2** and **6**, a reinforcing concave surface **133** is formed at a middle portion of an upper surface of the fork body **103**, and the reinforcing concave surface **133** is located at the lowest region of the fork body **103** and in smooth transitional connection with the two reinforcing arched strips **111**. The disposal of the reinforcing concave surface **133** further improves a deformation resistance of the fork body **103** under a force, thus facilitating transfer of a load from the prongs to the two reinforcing arched strips **111**.

(33) As shown in FIGS. **4** and **5**, a downwardly-recessed third reinforcing groove **114** is formed on the upper surface of the handle **101**, in each lateral edge portion of the handle **101**, one end of the third reinforcing groove **114** extends to an end of the handle **101** away from the first transition portion **102**, and the other end of the third reinforcing groove **114** extends through the first transition portion **102** to the lowest region of the fork body **103**, such that the two reinforcing arched strips **111** are disposed within the third reinforcing grooves **114**. In a sense, the third reinforcing grooves **114** of the handle **101** form the two curved lateral edge portions of the handle, respectively. The disposal of the two reinforcing arched strips **111** together with the third reinforcing groove **114** presents beautiful appearance and increases a connection strength between the reinforcing arched strips **111** and the handle **101** as well as bending strength and torsional strength of the handle **101**.

(34) A second transition portion **115** is formed at each of ends of the two reinforcing arched strips **111**, and the two reinforcing arched strips **111** are in smooth transitional connection with the third reinforcing groove **114** of the handle **101** through the second transition portions **115**. The disposal of the second transition portions **115** beautifies the external appearance of the handle **101**, facilitates transfer of a load at the top end of the reinforcing arched strips **111** and increases the structural strength of the end of the reinforcing arched strips **111**.

(35) A third transition portion **116** is formed at each of the ends of the two reinforcing arched strips **111** close to the first transition portion **102**, and the reinforcing arched strips **111** are in smooth transitional connection with the reinforcing concave surface **133** of the fork body **103** through the third transition portions **116**. The disposal of the third transition portions **116** beautifies the external appearance of the handle **101**, facilitates transfer of a load at the bottom end of the reinforcing arched strips **111** and increases the structural strength of the end of the reinforcing arched strips **111**.

(36) As shown in FIG. **5**, the two reinforcing arched strips **111** are connected with each other through a connection bridge **113**. One or more connection bridges **113** may be disposed. When one connection bridge **113** is disposed, the connection bridge **113** is disposed at an end of the handle **101** away from the first transition portion **102**, so as to improve the entire strength and the holding

comfort of the handle **101**. When two or more connection bridges **113** are disposed, the connection bridges **113** may be spaced apart. The connection bridge **113** and the two reinforcing arched strips **111** form an approximate plane to increase the holding convenience and comfort. Furthermore, information such as cavity position number may be printed thereon to facilitate quality inspection and mold replacement and the like during batch production.

(37) As shown in FIGS. **8** and **9**, the two reinforcing arched strips **111** are connected with each other through a fourth transition portion **117**, and the fourth transition portion **117** is a convex arc-surface structure and located at an end of the handle **101** away from the first transition portion **102**. The fourth transition portion **117** is connected smoothly between the two reinforcing arched strips **111** and the third reinforcing groove **114** of the handle **101**. Similar to the connection bridge **113**, the fourth transition portion **117** can not only increase the structural strength of the top end of the handle **101** but also improve the aesthetics and the holding comfort of the handle **101**.

(38) The upper surfaces of the two reinforcing arched strips **111** both are inclined surface structures with an angle of inclination being 1.4°-1.6°. A high end of the inclined surface structure is located at an end of the handle **101** away from the first transition portion **102**, and a low end of the inclined surface structure is located at an end of the handle **101** connected with the first transition portion **102**. The inclined surface structures of the reinforcing arched strips **111** increase the structural strength of the distal end of the handle **101** to ensure normal use of the handle **101** and also improve the holding comfort and the external aesthetics to some extent.

(39) The fast food fork body is an integral thin-walled structure with a thickness of 0.6-0.9 mm The undulated reinforcing structure ensures the entire structural strength of the thin-walled fast food fork body. The fast food fork body is used together with spoon and knife and has the same structure in handle as the spoon and knife. Therefore, the fast food fork body can be nested with the spoon and knife sequentially through the handle **101** to form a cutlery combination. The cutlery combination with integral thin-walled structure is smaller in volume and therefore widely applied in the fast food delivery of trains, airplanes and the like.

Embodiment 1-1

(40) As shown in FIGS. **1-6**, the handle **101**, the first transition portion **102** and the fork body **103** form the fast food fork body, and three prongs **131** are formed uniformly on the fork body **103**. The reinforcing concave surface **133** is formed at the middle portion of the fork body **103**. The second reinforcing groove **132** is formed at each of the three prongs **131**, the three second reinforcing grooves **132** are paralleled to each other, and the second reinforcing groove **132** located in middle penetrates through the reinforcing concave surface **133**.

(41) The third reinforcing groove **114** is formed downwardly along an edge of the upper surface of the handle **101** and extends from the top end of the handle **101** to the fork body **103**. Two reinforcing arched strips **111** with an angle of inclination being 1.5° are convexly disposed within the third reinforcing grooves **114**. The two reinforcing arched strips **111** are disposed in parallel and form the first reinforcing groove **112**. Further, the first reinforcing groove **112** may also be used a nesting groove to facilitate nesting packaging of the cutlery having the handle **101** of the same specifications. The second reinforcing groove **132** located in middle extends into the first reinforcing groove **112**. One connection bridge **103** is disposed at an upper part of the first reinforcing groove **112**. The connection bridge **103** is connected between the two reinforcing arched strips **111** and a cavity position number may be printed thereon. Ends of the two reinforcing arched strips **111** are in smooth transitional connection with the third reinforcing groove **114** through the second transition portions **115** respectively, and other ends of the two reinforcing arched strips **111** are in smooth transitional connection with the reinforcing concave surface **133** of the fork body **103** through the third transition portions **116** respectively.

(42) In this embodiment, an integral thin-walled structure with a thickness of 0.60 mm is formed through one-time solidification of a degradable paper pulp.

Embodiment 1-2

(43) As shown in FIGS. **7-10**, the handle **101**, the first transition portion **102** and the fork body **103** form the fast food fork body, and three prongs **131** are formed uniformly on the fork body **103**. The reinforcing concave surface **133** is formed on the middle portion of the fork body **103**. The second reinforcing groove **132** is formed on each of the three prongs **131**, the three second reinforcing grooves **132** are paralleled to each other, and the second reinforcing groove **132** located in middle penetrates through the reinforcing concave surface **133**.

Embodiment 2-1

(44) The third reinforcing groove **114** is formed downwardly along an edge of the upper surface of the handle **101** and extends from the top end of the handle **101** to the fork body **103**. Two reinforcing arched strips **111** with an angle of inclination being 1.6° are convexly disposed within the third reinforcing grooves **114**. The two reinforcing arched strips **111** are disposed in parallel and form the first reinforcing groove **112**. Further, the first reinforcing groove **112** may also be used a nesting groove to facilitate nesting packaging of the cutlery having the handle **101** of the same specifications. The second reinforcing groove **132** located in middle extends into the first reinforcing groove **112**. One connection bridge **103** is disposed at an upper part of the first reinforcing groove **112**. The connection bridge **103** is connected between the two reinforcing arched strips **111** and a cavity position number may be printed thereon. Ends of the two reinforcing arched strips **111** are in smooth transitional connection with the third reinforcing groove **114** through the fourth transition portions **117** respectively, and other ends of the two reinforcing arched strips **111** are in smooth transitional connection with the reinforcing concave surface **133** of the fork body **103** through the third transition portions **116** respectively.

(45) In this embodiment, an integral thin-walled structure with a thickness of 0.75 mm is formed through one-time solidification of a degradable paper pulp.

Spoon

Embodiment 2-1

(46) With reference to FIG. **11**, there is provided a disposable paper pulp fast food spoon, including a handle **201**, a transition portion **202** and a spoon body **203**. An end of the handle **201** is connected with an end of the spoon body **203** through the transition portion **202** to form a fast food spoon body.

(47) As shown in FIGS. **12-14**, the handle **201** has two curved lateral edge portions **201***a*. A U-shaped recess **211** is formed on an upper surface of the handle **201**, and the U-shaped recess **211** extends from an end of the handle **201** to the transition portion **202**; a double arch reinforcing portion or structure **204** is disposed in the U-shaped recess **211**, and an upper surface of the double arch reinforcing portion **204** is higher than the handle **201**; the double arch reinforcing portion **204** is formed by two paralleled arched strips **241** which are disposed along edges of two sides of the U-shaped recess **211** respectively, and a nesting groove **242** is formed between the two arched strips **241**. The groove-shaped structure of the nesting groove **242** can not only enable the handle **201** to have a nesting function but also increase a strength of the handle **201**. An undulant curved surface structure is formed between the two arched strips **241**, the nesting groove **242**, the U-shaped recess **211** and the handle **201**. An end of the arched strip **241** extends to an end of the U-shaped recess **211** and the other end of the arched strip **241** extends toward the lowest region of the spoon body **203** through the transition portion **202**. Preferably, the other end of the arched strip **241** extends to be close to but not over the lowest position of the spoon body **203**. Two parallel arched strips **241** can effectively improve the structural strength of the body of the fast food spoon, facilitate transfer of a load and enable the body of the fast food spoon to receive uniform force. The other end of the arched strip **241** may also extend slightly over the curved surface lowest position of the spoon body **203**, thereby ensuring the structural strength of the spoon body **203**. In other words, the handle **201** includes a first double arch reinforcing structure disposed between the two curved lateral edge portions **201***a* and extending along a length of the handle **201**. The transition portion **202** includes a second double arch reinforcing structure extending from the first double

arch reinforcing structure and toward the spoon body **203**. The spoon body includes a third double arch reinforcing structure extending from the second double arch reinforcing structure and toward a lowest region of the spoon body **203**. As shown in FIG. **11**, for example, the first double arch reinforcing structure, the second double arch reinforcing structure, and the third double arch reinforcing structure form a single one double arch reinforcing structure.

(48) The nesting groove **242** is disposed along a center axis of the handle **201**, a connection bridge **243** is disposed in the nesting groove **242**, and one or more connection bridges **243** are connected between the two arched strips **241** respectively. The disposal of one or more connection bridges **243** can further increase the structural strength of the handle **201**, thus avoiding twist of the handle **201**. Further, a plane structure or an approximate plane structure formed by the connection bridge **243** and the two arched strips **241** can effectively improve the gripping comfort and stability. In addition, the information such as cavity position number, trademark and patent number can be printed on a surface of the connection bridge **243**. The number of the connection bridges **243** may be determined based on a length of the handle **201**. If the handle **201** is short, for example, the handle of an ice cream spoon is short, the connection bridge **243** may not be disposed. If the handle **201** is long, for example, the handle of an ice spoon is long, one connection bridge **243** may be disposed or two to three connection bridges **243** may be disposed in a spacing, so as to ensure anti-bending and anti-twisting effect of the handle **201**.

(49) As shown in FIG. **15**, an upper surface of the arched strip **241** is an inclined surface structure. An angle of inclination of the inclined surface structure may be set to 1.0°-1.6° based on the length of the handle **201**. An inclined surface high end of the arched strip **241** is located at an end of the handle **201** away from the transition portion **202**, and an inclined surface low end of the arched strip **241** is located at an end of the handle **201** close to the transition portion **202**. The inclined surface structure of the arched strip **241** increases the structural strength of the distal end of the handle **201** to ensure normal use of the handle **201** and also improves the gripping comfort and external aesthetics to some extent.

(50) A first smooth transition portion **2411** is formed at the inclined surface high end of the arched strip **241**, and the inclined surface high end of the arched strip **241** is in smooth transitional connection with an end of the handle **201** through the first smooth transition portion **2411**. The disposal of the first smooth transition portion **2411** improves the structural aesthetics of the handle **201**, and a radian of the first smooth transition portion **2411** can convey a load better, thus facilitating increasing the overall strength of the fast food spoon.

(51) A second smooth transition portion **2412** is formed at the inclined surface low end of the arched strip **241** and is in smooth transitional connection with the spoon body **203** through the transition portion **202**. The second smooth transition portion **2412** extends to be close to the lowest region of the spoon body **203**. With disposal of the second smooth transition portion **2412**, the strength of the spoon body **203** can be increased and the structural aesthetics of the handle **201** can be improved. Furthermore, a radian of the second smooth transition portion **2412** can convey a load better so as to help increase the overall strength of the fast food spoon.

(52) A third smooth transition portion **244** is formed at an end of the double arch reinforcing portion **204**, and the inclined surface high ends of the two arched strip **241** are in smooth transitional connection with an end of the handle **201** through the third smooth transition portion **244**. With disposal of the third smooth transition portion **244**, the structural strength and aesthetics of the handle **201** can be improved and a good gripping comfort can also be provided. Furthermore, information such as cavity position number may be printed on a front or back face of the third smooth transition portion **244** to promote quality inspection and mold change and the like during mass production.

(53) The upper surface of the handle **201**, an upper surface of the transition portion **202** and an upper surface of the spoon body **203** are located basically in a same plane, such that an occupation space of the fast food spoon can be reduced on the basis of ensuring the strength of the fast food

spoon, helping packaging and transportation of fast foods as well as stacking of the fast food spoons.

(54) The handle **201**, the transition portion **202**, the spoon body **203** and the double arch reinforcing portion **204** are formed into an integral thin-walled fast food spoon through paper pulp solidification, thus facilitating mold opening and process operations.

(55) The integral thin-walled fast food spoon has a thickness of 0.6-0.9 mm Due to the small thickness, the volume and weight of the fast food spoon can be reduced and the production cost is lowered. With disposal of the reinforcing structure such as the double arch reinforcing portion **204**, it is guaranteed that the fast food spoon can still have good structural strength in a case of small material thickness and can be used normally.

(56) An end of the U-shaped recess **211** is in smooth transitional connection with the upper surface of the handle **201**, the other end of the U-shaped recess **211** is in smooth transitional connection with the transition potion **202**, and the transition portion **202** is in smooth transitional connection with the spoon body **203**. The smooth connection of the handle **201**, the transition portion **202** and the spoon body **203** helps to effectively convey a load.

(57) The disposable paper pulp fast food spoon according to the present disclosure may be used separately or together with other cutlery such as knife and fork. The handles of the knife and the fork may be manufactured into the same structure as the handle **201** of the present disclosure, such that the fast food spoon, the knife and the fork can be nested together in sequence through the handle **201** to form a disposable cutlery combination which features small volume and light weight and thus can be used in food delivery field of airplane, train and the like.

(58) As shown in FIG. **11**, the handle **201**, the transition portion **202** and the spoon body **203** form a fast food spoon body which has a total length of 135 mm The U-shaped recess **211** is formed downwardly along an edge on an upper surface of the handle **201**, a distance from the U-shaped recess **211** to the edge of the handle **201** is about 0.5 mm, and the U-shaped recess **211** extends to the transition portion **202**. The double arch reinforcing portion **204** formed by two arched strips **241** is formed on the U-shaped recess **211**, the two arched strips **241** are disposed in parallel along both sides of the U-shaped recess **211**, and the nesting groove **242** is formed between the two arched strips **241** to facilitate mutual nesting of the fast food spoons of the same specification. The upper surface of the two arched strips **241** is an inclined surface structure with an angle of inclination being 1.5°. The inclined surface high ends of the two arched strips **241** smoothly transition to an end of the U-shaped recess **211** through the first smooth transition portion **2411**, and the inclined surface low ends of the two arched strips **241** smoothly transition to the lowest region of the spoon body **203** through the second smooth transition portion **2412**.

(59) One connection bridge **243** is disposed in the nesting groove **242** and located at a side of the first smooth transition portion **2411**. An upper surface of the connection bridge **243** is lower than upper surfaces of the arched strips **241** where the connection bridge **243** is disposed. Information such as cavity position number may be printed on the connection bridge **243** to facilitate mass production. In addition, a height of the connection bridge **243** may be adjusted according to a nesting function requirement of the cutlery to control a nesting depth of the cutlery.

(60) The handle **201**, the transition portion **202**, the spoon body **203** and the double arch reinforcing portion **204** are formed into an integral thin-walled fast food spoon with a thickness of 0.75 mm by using a degradable paper pulp material through one-time solidification.

Embodiment 2-2

(61) As shown in FIG. **16**, the handle **201**, the transition portion **202** and the spoon body **203** form a fast food spoon body which has a total length of 89 mm The U-shaped recess **211** is formed downwardly along an edge on an upper surface of the handle **201**, a distance from the U-shaped recess **211** to the edge of the handle **201** is about 0.5 mm, and the U-shaped recess **211** extends to the transition portion **202**. The double arch reinforcing portion **204** formed by two arched strips **241** is formed on the U-shaped recess **211**, the two arched strips **241** are disposed in parallel along both

sides of the U-shaped recess **211**, and the nesting groove **242** is formed between the two arched strips **241** to facilitate mutual nesting of the fast food spoons of the same specification. The upper surface of the two arched strips **241** is an inclined surface structure with an angle of inclination being 1.6°. The inclined surface high ends of the two arched strips **241** smoothly transition to an end of the U-shaped recess **211** through the first smooth transition portion **2411**, and the inclined surface low ends of the two arched strips **241** smoothly transition to the lowest region of the spoon body **203** through the second smooth transition portion **2412**.

(62) The handle **201**, the transition portion **202**, the spoon body **203** and the double arch reinforcing portion **204** are formed into an integral thin-walled fast food spoon with a thickness of 0.6 mm by using a degradable paper pulp material through one-time solidification.

Embodiment 2-3

(63) As shown in FIG. **17**, the handle **201**, the transition portion **202** and the spoon body **203** form a fast food spoon body which has a total length of 210 mm The U-shaped recess **211** is formed downwardly along an edge on an upper surface of the handle **201**, a distance from the U-shaped recess **211** to the edge of the handle **201** is about 0.5 mm, and the U-shaped recess **211** extends to the transition portion **202**. The double arch reinforcing portion **204** formed by two arched strips **241** is formed on the U-shaped recess **211**, the two arched strips **241** are disposed in parallel along both sides of the U-shaped recess **211**, and the nesting groove **242** is formed between the two arched strips **241** to facilitate mutual nesting of the fast food spoons of the same specification. The upper surface of the two arched strips **241** is an inclined surface structure with an angle of inclination being 1.0°. The inclined surface high ends of the two arched strips **241** smoothly transition to an end of the U-shaped recess **211** through the third smooth transition portion **244**, and the inclined surface low ends of the two arched strips **241** smoothly transition to the lowest region of the spoon body **203** through the second smooth transition portion **2412**. The third smooth transition portion **244** is easy to grip, and the cavity position number may be printed on a front or back face of the third smooth transition portion **244** to facilitate mass production. Because the handle **201** of the fast food spoon body is long, three connection bridges **243** may be disposed in a spacing in the nesting groove **242** to prevent twist deformation of the handle **201**.

(64) The handle **201**, the transition portion **202**, the spoon body **203** and the double arch reinforcing portion **204** are formed into an integral thin-walled fast food spoon with a thickness of 0.6-0.9 mm by using a degradable paper pulp material through one-time solidification.

Knife

(65) As shown in FIGS. **18** and **23**, provided is a disposable paper pulp fast food knife, including a handle **301** and a knife body **302**. The knife body **302** is connected to an end of the handle **301**, and the handle **301** and the knife body **302** both are located in a same plane to form a fast food knife body.

(66) As shown in FIGS. **18-20**, **23** and **24**, a downwardly-recessed first groove body **303** is formed on an upper surface of the fast food knife body and extends from an end of the handle **301** to a middle portion of the knife body **302**. Two knife body reinforcing ribs **304** are convexly disposed in the first groove body **303** and extend from an edge of an end of the handle **301** to the knife body **302**. The two knife body reinforcing ribs **304** are disposed in parallel along a length direction of the handle **301**, and a downwardly-recessed reinforcing groove **341** is formed between the two knife body reinforcing ribs **304** as shown in FIGS. **21** and **25**. An undulated curved surface structure is formed by the handle **301**, the first groove body **303** on the handle **301**, the knife body reinforcing ribs **304** on the first groove body **303** and the reinforcing groove **341** between the two knife body reinforcing ribs **304**, which greatly increases the strength of the handle **301**, thus achieving better bending resistance and torsion resistance. A downwardly-recessed second groove body **305** is formed on an upper surface of the knife body **302**. One end of the second groove body **305** is in smooth transitional connection with the first groove body **303**, and the other end of the second groove body **305** extends to a tip of the knife body **302** along a length direction of the knife body

302. With the disposal of the second groove body **305** joining the first groove body **303**, a load is better transferred so that the load of the knife body **302** can be directly transferred to the handle **301** to ensure effective use of the knife body **302**. Even if a thin fast food knife body is made of paper pulp, the fast food knife body can still work normally.
(67) An end of the knife body reinforcing rib **304** away from a cutting edge extends to a connection position of the first groove body **303** and the second groove body **305**. In this way, a load of the knife body **302** can be transferred to the handle **301** along the second groove body **305**, the first groove body and the knife body reinforcing ribs **304**, thereby reducing occurrences of deformations of the knife body **302** and further guaranteeing effective use of the fast food knife body.
(68) As shown in FIGS. **20** and **24**, upper surfaces of the two knife body reinforcing ribs **304** are inclined surface structures higher than an upper surface of the handle **301**. A high end of the inclined surface structure is located at an end of the handle **301**, and a low end of the inclined surface structure is located on the knife body **302**. The two knife body reinforcing ribs **304** of inclined surface structure increase the holding comfort of the handle **301** and facilitate transfer of a load from the knife body **302** to an end of the handle **301**. Furthermore, its external appearance is made more beautiful.
(69) The inclined surface structures of the two knife body reinforcing ribs **304** have an angle of inclination of 1.4°-1.6° and preferably 1.5°, which is helpful to mold opening and process operations.
(70) A first transition portion **342** is formed at the high ends of the inclined surface structures of the two knife body reinforcing ribs **304**, and a second transition portion **343** is formed at the low ends of the inclined surface structures of the two knife body reinforcing ribs **304**. The knife body reinforcing ribs **304** are in smooth transitional connection with the first groove body **303** through the first transition portion **342**, and the knife body reinforcing ribs **304** are in smooth transitional connection with the knife body **302** through the second transition portion **343**. With disposal of the first transition portion **342** and the second transition portion **343**, the connections of the two knife body reinforcing ribs **304** and the handle **301**/the knife body **302** are made smoother and more beautiful, and transfer of the load is facilitated. In this way, the structural strength of both ends of the knife body reinforcing ribs is significantly increased.
(71) The first transition portion **342** includes two strip-shaped first unit transition portions. Each of the knife body reinforcing ribs **304** is connected with the first unit transition portion to form an elongated reinforcing rib parallel to the length direction of the handle **301**. Forming the integral elongated structure by joining the first unit transition portion to the knife body reinforcing rib **304** guarantees direct transfer of a load. Further, the structural strength and the aesthetics of the handle **301** are improved.
(72) As shown in FIG. **22**, a connection bridge **344** is disposed between the two elongated reinforcing ribs and located in the reinforcing groove **341**. The disposal of the connection bridge **344** further increases the structural strength of the handle **301**, especially the torsion resistance and improves the holding comfort. Further, information such as cavity position number may be printed on a surface of the connection bridge **344** to facilitate quality inspection and mold replacement and the like during batch production. Preferably, one connection bridge **344** is disposed. Alternatively, a plurality of connection bridges **344** may be disposed according to the length of the handle **301** and the design requirements. When a plurality of connection bridges **344** are disposed, these connection bridges **344** are spaced apart.
(73) As shown in FIG. **26**, the first transition portion **342** is an arc-surface structure such that the first transition portion **342** is in smooth transitional connection with the two knife body reinforcing ribs **304** and the first groove body **303**. An arc-shaped holding structure is formed between the knife body reinforcing ribs **304** and the handle **301** by use of the first transition portion **342** of arc surface structure, so as to improve the holding comfort while ensuring the top strength of the handle **301**.

(74) The second transition portion **343** includes two strip-shaped second unit transition portions. Each of the knife body reinforcing ribs **304** is connected with the second unit transition portion to form an elongated reinforcing rib parallel to the length direction of the handle **301**. One elongated reinforcing rib is in smooth transitional connection with the first groove body **303** and the other elongated reinforcing rib is in smooth transitional connection with the first groove body **303** and the second groove body **305**. Forming the integral elongated structure by joining the second unit transition portion to the knife body reinforcing rib **304** guarantees direct transfer of a load. Further, the structural strength and the aesthetics of the handle **301** are improved.
(75) The disposable paper pulp fast food knife is an integral thin-walled structure formed through paper pulp solidification, and has a thickness of 0.6-0.9 mm With the undulated reinforcing structure, the structural strength of the handle **301** is guaranteed such that the handle **301** can be still effectively used in a case of small thickness, thereby reducing production costs.
(76) The disposable paper pulp fast food knife is used together with spoon and fork and has the same structure in handle as the spoon and fork. Therefore, the disposable paper pulp fast food knife can be nested with the spoon and fork sequentially through the handle **301** to form a cutlery combination. The cutlery combination is smaller in volume and therefore widely applied in the fast food delivery of trains, airplanes and the like.
Embodiment 3-1
(77) As shown in FIGS. **18-22**, the handle **301** and the knife body **302** form the fast food knife body. The first groove body **303** is formed on the fast food knife body and extends from an edge of an end of the handle **301** to a middle portion of the knife body **302**. Two knife body reinforcing ribs **304** are disposed in parallel in the first groove body **303** along the length direction of the handle **301**. The reinforcing groove **341** is formed between the two knife body reinforcing ribs **304**. The reinforcing groove **341** may also be used as a nesting groove to facilitate nesting of the fast food knife bodies of the same specifications. The second groove body **305** is formed on the knife body **302** along the length direction of the knife body **302** and extends from a tip of the knife body **302** to the first groove body **303**. The first groove body **303** and the second groove body **305** connect with one of the knife body reinforcing ribs **304** so as to enable a load on the knife body **302** to be effectively transferred to the handle **301**.
(78) The two knife body reinforcing ribs **304** have an angle of inclination of 1.5°. Ends of the two knife body reinforcing ribs **304** are in smooth connection with the first groove body **303** through the strip-shaped first transition portion **342**, where the other end of one of the two knife body reinforcing ribs **304** is in smooth transitional connection with the first groove body **303** through the second transition portion **343**, and the other end of the other of the two knife body reinforcing ribs **304** is in smooth transitional connection with a connection position of the first groove body **303** and the second groove body **305** through the second transition portion **343**. One connection bridge **344** is disposed in the reinforcing groove **341** between the two knife body reinforcing ribs **304**. The connection bridge **344** is disposed close to the first transition portion **342**. With disposal of the two knife body reinforcing ribs **304**, the first transition portion **342**, the second transition portion **343**, and the connection bridge **344**, the fast food knife body has good bending resistance and torsion resistance.
(79) In this embodiment, the disposable paper pulp fast food knife is integrally formed through paper pulp solidification, and has a thin-walled structure with a thickness of 0.6 mm
Embodiment 3-2
(80) As shown in FIGS. **23-26**, the handle **301** and the knife body **302** form the fast food knife body. The first groove body **303** is formed on the fast food knife body and extends from an edge of an end of the handle **301** to a middle portion of the knife body **302**. Two knife body reinforcing ribs **304** are disposed in parallel in the first groove body **303** along the length direction of the handle **301**. The reinforcing groove **341** is formed between the two knife body reinforcing ribs **304**. The reinforcing groove **341** may also be used as a nesting groove to facilitate nesting of the fast food

knife bodies of the same specifications. The second groove body **305** is formed on the knife body **302** along the length direction of the knife body **302** and extends from a tip of the knife body **302** to the first groove body **303**. The first groove body **303** and the second groove body **305** connect with one of the knife body reinforcing ribs **304** so as to enable a load on the knife body **302** to be effectively transferred to the handle **301**.

(81) The two knife body reinforcing ribs **304** have an angle of inclination of 1.4°. Ends of the two knife body reinforcing ribs **304** are in smooth connection with the first groove body **303** through the first transition portion **342** of arc-surface structure, where the other end of one of the two knife body reinforcing ribs **304** is in smooth transitional connection with the first groove body **303** through the second unit transition portion, and the other end of the other of the two knife body reinforcing ribs **304** is in smooth transitional connection with a connection position of the first groove body **303** and the second groove body **305** through another second unit transition portion. With disposal of the two knife body reinforcing ribs **304**, the first transition portion **342** and the second transition portion **343**, the fast food knife body has good bending resistance and torsion resistance.

(82) In this embodiment, the disposable paper pulp fast food knife is integrally formed through paper pulp solidification, and has a thin-walled structure with a thickness of 0.6-0.9 mm.

(83) The above descriptions are merely illustrative of preferred embodiments of the present disclosure rather than limiting of the scope of protection of the present disclosure. Therefore, any modifications, equivalent substitutions and improvements and the like made within the spirit and principle of the present disclosure shall all fall within the scope of protection of the present disclosure.

Claims

1. A pulp spoon comprising: a handle; a spoon body; and a transition portion by which the handle is connected with the spoon body, wherein the handle comprising two curved lateral edge portions, and a first double arch reinforcing structure disposed between the two curved lateral edge portions and extending along a length of the handle, an upper surface of the first double arch reinforcing structure being higher than the two curved lateral edge portions when the pulp spoon is in an upright position, wherein the transition portion comprising a second double arch reinforcing structure which extends from the first double arch reinforcing structure and toward the spoon body, wherein the spoon body comprising a third double arch reinforcing structure which extends from the second double arch reinforcing structure and toward a lowest region of the spoon body, and wherein the first double arch reinforcing structure, the second double arch reinforcing structure, and the third double arch reinforcing structure define a single one double arch reinforcing structure.
2. The pulp spoon of claim 1, wherein the first double arch reinforcing structure comprises two parallel arched strips and a nesting groove therebetween.
3. The pulp spoon of claim 2, wherein the nesting groove is higher than the two curved lateral edge portions.
4. The pulp spoon of claim 2, wherein the first double arch reinforcing structure further comprises at least one connection bridge in the nesting groove that connects the two parallel arched strips with each other.
5. The pulp spoon of claim 2, wherein the nesting groove is centrally disposed between the two curved lateral edge portions.
6. The pulp spoon of claim 5, wherein the nesting groove is disposed higher than the two curved lateral edge portions.
7. The pulp spoon of claim 1, wherein the handle, the spoon body and the transition portion form an integral structure.
8. A piece of pulp cutlery, comprising: a handle; a working body; and a transition portion by which

the handle is connected with the working body, wherein the handle comprising two curved lateral edge portions, and a first double arch reinforcing structure disposed between the two curved lateral edge portions and extending along a length of the handle, an upper surface of the first double arch reinforcing structure being higher than the two curved lateral edge portions when the piece of pulp cutlery is in an upright position, wherein the transition portion comprising a second double arch reinforcing structure which extends from the first double arch reinforcing structure and toward the working body, and wherein the working body comprising a third double arch reinforcing structure which extends from the second double arch reinforcing structure and toward a lowest region of the working body, and wherein the first double arch reinforcing structure, the second double arch reinforcing structure, and the third double arch reinforcing structure define a single one double arch reinforcing structure.

9. The piece of pulp cutlery of claim 8, wherein the first double arch reinforcing structure comprises two parallel arched strips and a nesting groove therebetween.

10. The piece of pulp cutlery of claim 9, wherein the nesting groove is higher than the two curved lateral edge portions.

11. The piece of pulp cutlery of claim 9, wherein the first double arch reinforcing structure further comprises at least one connection bridge in the nesting groove that connects the two parallel arched strips with each other.

12. The piece of pulp cutlery of claim 8, wherein the handle, the working body and the transition portion form an integral structure.

13. The piece of pulp cutlery of claim 8, wherein the working body comprises one selected from the group consisting of a fork body, and a spoon body.